AE-592: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)

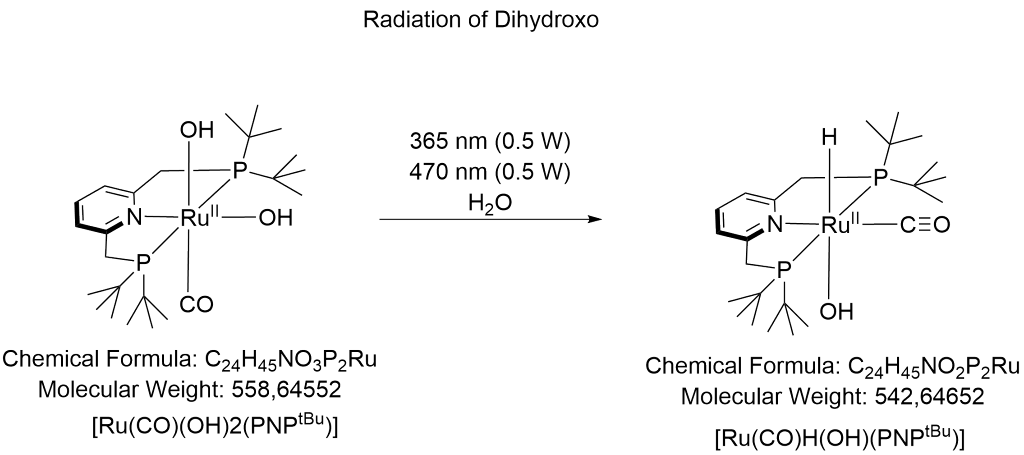
Date: 2025-10-14
Tags: Dihydroxo PNP P(tBu)N(py)P(tBu) [Ru(CO)(OH)2(PNP)] NMR AE Future 1H 31P Dual irradiation two-photon photocatalysis
Category: Two-photon
Status: Done
Created by: Alexander Eith

Objective

Testing dual irradiation effect of [Ru(CO)(OH)2(PNP)] at higher irradiance, to get better conversion after 16 h

Reproduction of AE-590

Reaction scheme/sample structure



ChemDraw File (linked): AE-590.cdxml

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	Two-photon - AE-590: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)
Similar experiments	Two-photon - AE-519: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) Two-photon - AE-511: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) - reproduction II Two-photon - AE-502: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) - reproduction Two-photon - AE-500: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1)

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	Concentration [mM]

[Ru(CO)(OH)2(PNP{tBu})]	Two-photon - AE-591: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water	/	/	/	/	/	558.65	/	3 * 0.6	concentrated (approx. 1.44 mM)

Irradiation Parameters

Power measurement was performed in experiment Two-photon - AE-590: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)

	Name
Used Set-up	Equipment - Advanced irradiation setup V1.0 I
Irradiation set-up number	Equipment - Irradiation setup 2 (CEEC II, E004)

	Light Source Name	**Used power source**	**Wavelength [nm]**	**Power Setting [mW]**	**Analogue Setting [-]**
First light source	Light Source - UHP LED 365 nm-2	Power Sources - BLS-18000-1 1	365	500	9.68
Second light source	Light Source - UHP LED 470 nm	Power Sources - BLS-13000-1E	470	600	8.94

AE-592-1

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Used light source	365 + 470 nm
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration [h:min]	17:32

AE-592-2

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Used light source	365 nm
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration [h:min]	17:30

AE-592-3

Used beam combiner [Name or None]	Beam Combiner - Beam Combiner LCS-BC25-0409 -1
Used light source	470 nm
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	/
Irradiation duration [h:min]	17:28

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using either standard Protocol - Schlenk Technique

The synthesis/reactions were performed in the dark by either wrapping the reaction flasks in aluminum foil or turning the lab light and light in the fume hood off, when the flasks were uncovered.

Date	Time	Step	Observations	Pictures
13.10	16:05	Two Young type NMR tube was prepared according to Protocol - Preparation of NMR Sample (from step 3) using 0.65 mL of **AE-591**	**AE-592-1** and **AE-592-2**	/

		From **AE-592-1** an NMR was measured (**AE-592-1-1**)	**AE-592-1-1,** slight yellow soultion, maybe very small amount of solid deposited on the wall	AE-592-1-1.jpg
		From **AE-592-2** an NMR was measured (**AE-592-2-1**)	**AE-592-2-1**, slight yellow soultion	AE-592-2-1.jpg
14.10	17:14	The power was adjusted using Power Meter - 843-R-USB + 919P-020-12 in Equipment - Advanced power measurment setup V1.0 (365 nm: 500 mW, 470 nm: 500 mW)	/	/
	17:19	**AE-590-1** was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	back before irrad 1.jpg
	17:20	The irradiaiton of **AE-592-1** was started	/	back during irrad 1.jpg
	10:52	The irradiaiton of **AE-592-1** was stopped	/	back after irrad 1.jpg
	12:42	From **AE-592-1** an NMR was measured (**AE-592-1-2**)	**AE-592-1-2**, darker orange, some solids formed on the walls	AE-592-1-2.jpg
	14:20	One Young type NMR tube was prepared according to Protocol - Preparation of NMR Sample (from step 3) using 0.65 mL of **AE-591** using a Prep work - AE-489: Capillary production	**AE-592-3**	/
	14:50	From **AE-592-3** an NMR was measured (**AE-592-3-1**)	**AE-592-3-1**, slight yellow soultion	AE-592-3-1.jpg
	16:41	**AE-592-2** was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	back before irrad 2.jpg
	16:42	The irradiaiton of **AE-592-2** was started	/	back during irad 2.jpg
	10:12	The irradiaiton of **AE-592-2** was stopped	no significant change	back after irrad 2.jpg
	10:43	From **AE-592-2** an NMR was measured (**AE-592-2-2**)	**AE-592-2-2**, no change	AE-592-2-2.jpg
	15:05	**AE-592-3** was placed in the setup using the Photoreactor - AE-558: Design and print of holder and holder disk for 2 NMR tubes Holder_2NMR_tubes	no change	back before irrad 3.jpg

	15:10	The irradiaiton of **AE-592-3** was started	/	back during irrad 3.jpg
	08:38	The irradiaiton of **AE-592-3** was stopped	no significant change	back after irrad 3.jpg
	09:53	From **AE-592-3** an NMR was measured (**AE-592-3-2**)	**AE-592-3-2**, no change	AE-592-3-2.jpg

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation
14.10	11:22	AE-592-1-1	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-592-1-1.zip	/	AE-592-1-1_10.nmrium	Two-photon - AE-590: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)	As expected, before irrad, reactant, and minor impurtiy
15.10	16:27	AE-592-1-2	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-592-1-2.zip	/	AE-592-1-1_10.nmrium	Two-photon - AE-590: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)	Mainly as expected: formation of hydride species and side products, but more side products as expected
14.10	15:47	AE-592-2-1	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-592-2-1.zip	/	AE-592-1-1_10.nmrium	/	As expected, before irrad, reactant, and minor impurtiy
16.10	14:17	AE-592-2-2	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-592-2-2.zip	/	AE-592-1-1_10.nmrium	/	No conversion observed
16.10	02:02	AE-592-3-1	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-592-3-1.zip	/	AE-592-1-1_10.nmrium	Two-photon - AE-500: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1)	As expected, before irrad, reactant, and minor impurtiy
17.10	14:32	AE-592-3-2	1H, 31P{1H}quant NMR	IAAC 400 MHz	H2O	AE-592-3-2.zip	/	AE-592-1-1_10.nmrium	Two-photon - AE-500: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1)	As expected, no conversion observed

Results

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product, 57 ppm) [%]	Yield (soluable side products) [%]	Yield (other side products) [%]	Data
AE-592-1	17	0	7	41	20	15	AE-592.xlsx
AE-592-3	0	0	88	0	3	9	AE-592.xlsx

AE-592-2: no evaluated, since LED was connected to wrong port of the beam combiner

Worked in general, good agreement of AE-592-1 with AE-590-1 for hydride yield, but more sideproducts formed

AE-592-3: as expected no hydride formation

Future recommendations

Old procedure	Problem	Suggested new procedure
LEDs connected to the wrong ports of the beam combiner for irrad with 365 nm	way less power output than desired	do connect them correctly

Linked experiments

Two-photon - AE-500: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1)

Two-photon - AE-502: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) - reproduction

Two-photon - AE-511: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1) - reproduction II

Two-photon - AE-519: Dual Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-456-1)

Two-photon - AE-581: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water

Two-photon - AE-590: Dual irradiation of [Ru(CO)(OH)2(PNP)] (AE-570)

Two-photon - AE-591: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water

Linked resources

Beam Combiner - Beam Combiner LCS-BC25-0409 -1

Equipment - Advanced irradiation setup V1.0 I

Equipment - Advanced power measurment setup V1.0 I

Equipment - Manual irradiation setup

Equipment - Irradiation setup 2 (CEEC II, E004)

Light Source - UHP LED 470 nm-1

Light Source - UHP LED 365 nm-2

Power Meter - 843-R-USB + 919P-020-12

Power Sources - BLS-13000-1E

Power Sources - BLS-18000-1 1

Attached files

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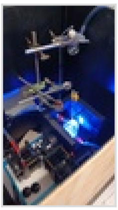

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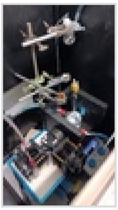

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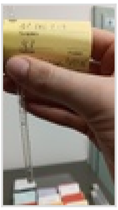

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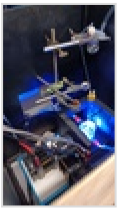

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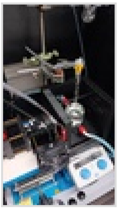

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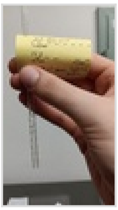

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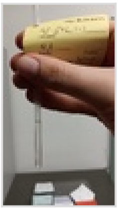

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